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APPLICATION NO.	FILING DATE	FIRST NAMED INVENTOR	ATTORNEY DOCKET NO.	CONFIRMATION NO.
17/570,917	01/07/2022	Adrian I. Bot	K-1104-US-NP	9051

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EXAMINER	
XIAO, YAN	

ART UNIT	PAPER NUMBER
1642	

NOTIFICATION DATE	DELIVERY MODE
12/13/2024	ELECTRONIC

Please find below and/or attached an Office communication concerning this application or proceeding.

The time period for reply, if any, is set in the attached communication.

Notice of the Office communication was sent electronically on above-indicated "Notification Date" to the following e-mail address(es):

IPUnpublished@gilead.com

Office Action Summary	Application No. 17/570,917	Applicant(s) Bot et al.	
	Examiner YAN XIAO	Art Unit 1642	AIA (FITF) Status Yes

-- The MAILING DATE of this communication appears on the cover sheet with the correspondence address --

Period for Reply

A SHORTENED STATUTORY PERIOD FOR REPLY IS SET TO EXPIRE 3 MONTHS FROM THE MAILING DATE OF THIS COMMUNICATION.

- Extensions of time may be available under the provisions of 37 CFR 1.136(a). In no event, however, may a reply be timely filed after SIX (6) MONTHS from the mailing date of this communication.
- If NO period for reply is specified above, the maximum statutory period will apply and will expire SIX (6) MONTHS from the mailing date of this communication.
- Failure to reply within the set or extended period for reply will, by statute, cause the application to become ABANDONED (35 U.S.C. § 133). Any reply received by the Office later than three months after the mailing date of this communication, even if timely filed, may reduce any earned patent term adjustment. See 37 CFR 1.704(b).

Status

1)☑ Responsive to communication(s) filed on 09/05/2024.

☐ A declaration(s)/affidavit(s) under **37 CFR 1.130(b)** was/were filed on _____.

2a)☑ This action is **FINAL.** 2b) ☐ This action is non-final.

3)☐ An election was made by the applicant in response to a restriction requirement set forth during the interview on _____; the restriction requirement and election have been incorporated into this action.

4)☐ Since this application is in condition for allowance except for formal matters, prosecution as to the merits is closed in accordance with the practice under *Ex parte Quayle*, 1935 C.D. 11, 453 O.G. 213.

Disposition of Claims*

5) ☑ Claim(s) 1-18 is/are pending in the application.

5a) Of the above claim(s) 3-8 and 12-17 is/are withdrawn from consideration.

6) ☐ Claim(s) _____ is/are allowed.

7) ☑ Claim(s) 1-2,9-11 and 18 is/are rejected.

8) ☐ Claim(s) _____ is/are objected to.

9) ☐ Claim(s) _____ are subject to restriction and/or election requirement

* If any claims have been determined allowable, you may be eligible to benefit from the **Patent Prosecution Highway** program at a participating intellectual property office for the corresponding application. For more information, please see http://www.uspto.gov/patents/init_events/pph/index.jsp or send an inquiry to **PPHfeedback@uspto.gov.**

Application Papers

10)☐ The specification is objected to by the Examiner.

11)☐ The drawing(s) filed on _____ is/are: a)☐ accepted or b)☐ objected to by the Examiner.

Applicant may not request that any objection to the drawing(s) be held in abeyance. See 37 CFR 1.85(a).

Replacement drawing sheet(s) including the correction is required if the drawing(s) is objected to. See 37 CFR 1.121(d).

Priority under 35 U.S.C. § 119

12)☐ Acknowledgment is made of a claim for foreign priority under 35 U.S.C. § 119(a)-(d) or (f).

Certified copies:

a)☐ All b)☐ Some** c)☐ None of the:

1.☐ Certified copies of the priority documents have been received.

2.☐ Certified copies of the priority documents have been received in Application No. _____.

3.☐ Copies of the certified copies of the priority documents have been received in this National Stage application from the International Bureau (PCT Rule 17.2(a)).

** See the attached detailed Office action for a list of the certified copies not received.

Attachment(s)

1) ☐ Notice of References Cited (PTO-892)

2) ☐ Information Disclosure Statement(s) (PTO/SB/08a and/or PTO/SB/08b) Paper No(s)/Mail Date _____.

3) ☐ Interview Summary (PTO-413) Paper No(s)/Mail Date _____.

4) ☐ Other: _____.

Notice of Pre-AIA or AIA Status

1. The present application, filed on or after March 16, 2013, is being examined under the first inventor to file provisions of the AIA.

DETAILED ACTION

2. The amendment/argument filed 09/05/2024 is acknowledged and has been entered.

Claims 1-2 and 9-11 have been amended. New claim 18 has been added.

3. Claims 1-18 are pending in the application. Claims 3-8 and 12-17 have been withdrawn from further consideration pursuant to 37 CFR 1.142(b) as being drawn to nonelected inventions, there being no allowable generic or linking claim. Election was made **without** traverse in the reply filed on 02/22/2024.

4. Claims 1-2, 9-11 and 18 have been examined.

Grounds of Objection and Rejection Withdrawn

5. Unless specifically reiterated below, Applicant's amendment and/or arguments have obviated or rendered moot the grounds of objection and rejection set forth in the previous Office action mailed 04/05/2024.

New Grounds of Rejection

Claim Rejections - 35 USC § 102

6. The following is a quotation of the appropriate paragraphs of 35 U.S.C. 102 that form the basis for the rejections under this section made in this Office action:

> A person shall be entitled to a patent unless –
>
> (a)(1) the claimed invention was patented, described in a printed publication, or in public use, on sale or otherwise available to the public before the effective filing date of the claimed invention.

7. Claims 1-2, 11 and 18 are rejected under 35 U.S.C. 102(1) as anticipated by PRUTEANU-MALINICI et al (WO2019213282 published 07 November 2019, IDS), or in

the alternative, under 35 U.S.C. 103 as obvious over PRUTEANU-MALINICI et al (WO2019213282 published 07 November 2019, IDS).

Claims 1-2, 11 and 18 are herein drawn to a method for treating a hematological cancer in a patient comprising: measuring a level of CD27+CD28+ naive CD4+ T helper (Th) cells in an apheresis product from said patient; and administering an effective dose of T cells comprising a chimeric antigen receptor if the level of CD27+CD28+ naive CD4+ Th cells is over a cut-off percentage value measured as a percentage of total leukocytes, or administering an effective dose of T cells comprising the chimeric antigen receptor and a combination therapy if the level of CD27+CD28+ naive CD4+ Th cells is below said cut-off percentage value wherein said cut-off percentage value is between 0.1% - 0.5%, and wherein chimeric antigen receptor comprises an anti-CD19 single chain variable fragment (scFv).

PRUTEANU-MALINICI et al. teach a method of evaluating or monitoring the effectiveness of a CAR-expressing cell (e.g., T cell, NK cell) therapy (e.g., a CD19 CAR-T cell) in a subject having cancer (e.g., a hematological cancer), comprises a measure of a combination of one or more of a CD27 biomarker, a CD45RO biomarker, a CCR7 biomarker, a HLA-DR biomarker, a CD95 biomarker, a CD127 biomarker, a CD4 biomarker, a CD8 biomarker, a TH1+helper T cell gene set signature, a TH2+ helper T cell gene set signature, a memory T cell (e.g., a CD8+ memory T cell, e.g., a naive T cell (TN), e.g. a memory stem cell (TSCM), e.g. a central memory T cell (TCM), e.g. an effector memory T cell (TEM)) gene set signature, and a CD19CAR-expressing cell gene set signature; see entire document, e.g., pages 14, 76-77 and 81.

PRUTEANU-MALINICI et al. teach naive T cell (TN) markers include CD27+, CD28+, see pages 66-67.

Thus, the method of PRUTEANU-MALINICI et al. comprises measuring CD27+, CD28+, CD4+, helper T cell.

For claim 2, PRUTEANU-MALINICI et al. teach determining a cut-off value; see page 33-lines 19-24, page 82-lines 9-18.

For claims 11 and 18, PRUTEANU-MALINICI et al. teach a combination therapy of CD19 CAR-T cell with additional therapeutic agent (e.g., Cyclophosphamide, Doxorubicin); see pages 71, 234-238, 262, Example 4.

Although PRUTEANU-MALINICI et al. teach a measure of a combination of biomarkers set forth above; however, PRUTEANU-MALINICI et al. do not expressly state only measure the combination of CD27+CD28+ naive CD4+ T helper cell.

Given that PRUTEANU-MALINICI et al. teach measuring a combination of biomarkers which include CD27+, CD28+, CD4+ and T helper cell; it would have been *prima facie* obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to combine the teachings of the reference so as to measure CD27+CD28+ naive CD4+ T helper cell to arrive the claimed invention, because some teachings, suggestion, or motivation in the prior art that would lead to one of ordinary skill to modify the prior art reference or to combine prior art reference teachings to arrive at the claimed invention.

Claim Rejections - 35 USC § 103

8. The following is a quotation of 35 U.S.C. 103 which forms the basis for all obviousness rejections set forth in this Office action:

> A patent for a claimed invention may not be obtained, notwithstanding that the claimed invention is not identically disclosed as set forth in section 102 of this title, if the differences between the claimed invention and the prior art are such that the claimed invention as a whole would have been obvious before the effective filing date of the claimed invention to a person having ordinary skill in the art to which the claimed invention pertains. Patentability shall not be negated by the manner in which the invention was made.

9. The factual inquiries set forth in *Graham v. John Deere Co.*, 383 U.S. 1, 148 USPQ 459 (1966), that are applied for establishing a background for determining obviousness under 35 U.S.C. 103 are summarized as follows:

1. Determining the scope and contents of the prior art.

2. Ascertaining the differences between the prior art and the claims at issue.

3. Resolving the level of ordinary skill in the pertinent art.

4. Considering objective evidence present in the application indicating obviousness or nonobviousness.

10. Claims 1 and 9-10 are rejected under 35 U.S.C. 103 as being unpatentable over PRUTEANU-MALINICI et al (WO2019213282 published 07 November 2019, IDS) in

view of Wang et al (Oncotarget, Vol. 8, (No.3), published December 27, 2016, pgs. 5414-5425) (of record).

Claims 9-10 are herein drawn to the method of claim 1, further comprising: measuring a lymphocyte to monocyte ratio in a baseline hematology count of said patient.

The teachings of PRUTEANU-MALINICI et al. have been set forth in the above rejection of claims 1-2, 11 and 18 under 35 U.S.C. 102(1).

PRUTEANU-MALINICI et al. do not teach measuring a lymphocyte to monocyte ratio.

However, this deficiency is remedied by Wang et al.

Wang et al. teach lymphocyte to monocyte ratio is associated with prognosis of diffuse large B-cell lymphoma (DLBCL); see entire document, e.g., title, abstract. Wang et al. teach lymphocyte to monocyte ratio (LMR) play an essential role in predicting clinical prognosis of DLBCL; see third paragraph-left col. of page 5422. Wang et al. teach cut-off values for lymphocyte to monocyte ratio (LMR); see right col. of page 5415.

It would have been *prima facie* obvious to one of ordinary skill in the art before the effective filing date of the claimed invention to combine the teachings of the references so as to include measuring lymphocyte to monocyte ratio in the method of PRUTEANU-MALINICI et al. to arrive the claimed invention. One would have been motivated to do so because PRUTEANU-MALINICI et al. teach a method of evaluating or monitoring the effectiveness of a CAR-expressing cell (e.g., T cell, NK cell) therapy (e.g., a CD19 CAR-T cell) in a subject having cancer (e.g., a hematological cancer), comprises a measure of a combination of one or more of CD27+, CD28+, CD4+ and T helper cell; Wang et al. teach lymphocyte to monocyte ratio is associated with prognosis of diffuse large B-cell lymphoma (DLBCL). Thus, one of ordinary skill in the art would have a reasonable expectation of success that by combining the teachings of the references so as to include measuring lymphocyte to monocyte ratio in the method of PRUTEANU-MALINICI et al. to arrive the claimed invention, because lymphocyte to monocyte ratio (LMR) play an essential role in predicting clinical prognosis of hematological cancer as taught by Wang et al.

Conclusion

11. No claim is allowed.

12. The After Final Consideration Pilot 2.0 (AFCP 2.0) has been extended through December 14, 2024. The extension of AFCP 2.0 is part of the USPTO's on-going efforts towards compact prosecution and increased collaboration between examiners and stakeholders. To be eligible for consideration under AFCP 2.0, you must file a response under 37 CFR §1.116, which includes a request for consideration under the pilot (Form PTO/SB/434) and an amendment to at least one independent claim that does not broaden the scope of the independent claim in any aspect.

13. Applicant's amendment necessitated the new ground(s) of objection/rejection presented in this Office action. Accordingly, **THIS ACTION IS MADE FINAL**. See MPEP § 706.07(a). Applicant is reminded of the extension of time policy as set forth in 37 CFR 1.136(a).

A shortened statutory period for reply to this final action is set to expire THREE MONTHS from the mailing date of this action. In the event a first reply is filed within TWO MONTHS of the mailing date of this final action and the advisory action is not mailed until after the end of the THREE-MONTH shortened statutory period, then the shortened statutory period will expire on the date the advisory action is mailed, and any extension fee pursuant to 37 CFR 1.136(a) will be calculated from the mailing date of the advisory action. In no event, however, will the statutory period for reply expire later than SIX MONTHS from the date of this final action.

14. Any inquiry concerning this communication or earlier communications from the examiner should be directed to YAN XIAO whose telephone number is (571)270-3578. The examiner can normally be reached M-F 8-5 EST.

Examiner interviews are available via telephone, in-person, and video conferencing using a USPTO supplied web-based collaboration tool. To schedule an

interview, applicant is encouraged to use the USPTO Automated Interview Request (AIR) at http://www.uspto.gov/interviewpractice.

If attempts to reach the examiner by telephone are unsuccessful, the examiner's supervisor, Samira Jean-Louis can be reached on 571-270-3503. The fax phone number for the organization where this application or proceeding is assigned is 571-273-8300.

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/YAN XIAO/
Primary Examiner, Art Unit 1642